

Initial Setup for the ETL process

This document covers the initial setup requirements before creating and running AWS Glue ETL jobs and also showcases the steps to be followed before setting up ETL process for a new Vendor's feed data.

Note: This document does not cover other related services mentioned nor it covers information regarding role creations. For these additional links are mentioned below:

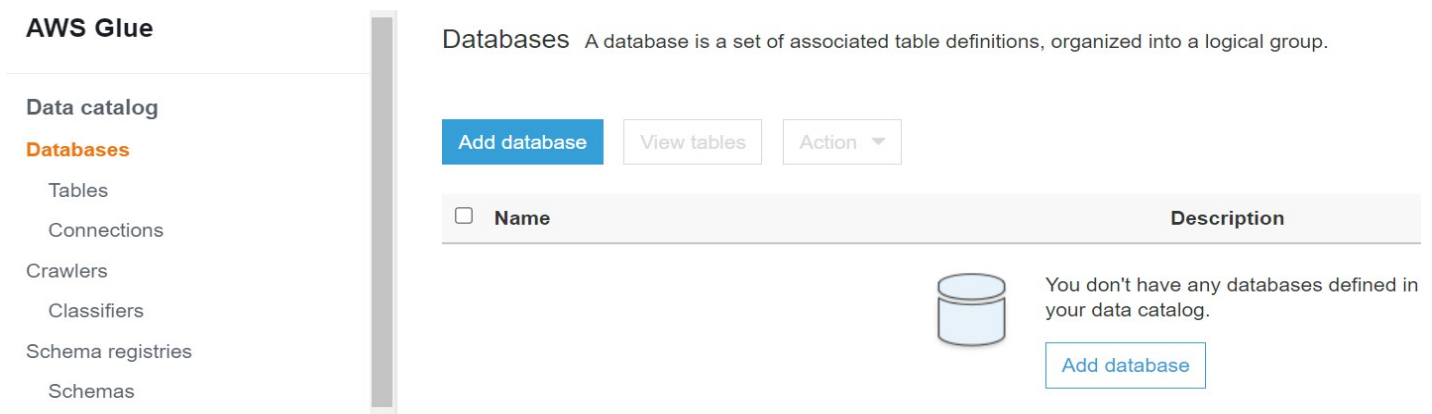
1. Initial setup steps

1. Database creation for centralized PostgreSQL database in AWS Glue Catalog
2. Connection setup for centralized PostgreSQL database
3. Crawler creation to create relevant tables in the Glue Catalog and get the schema and metadata information from the PostgreSQL database

2. One timer steps for Vendor's feed data

1. Database creation for the Vendor's feed folder in AWS Glue Catalog
2. Crawler creation to create relevant table in Glue Catalog and get the schema and metadata information from the feed file

1.1 Database creation for PostgreSQL in Glue Catalog:



- Click on Add database button
- And enter a unique and relevant name for the database
- Add description (optional)
- Click on create button

1.2 Creating connection to connect with RDS PostgreSQL database instance:

AWS Glue

Data catalog

Databases

Tables

Connections

Crawlers

Classifiers

Schema registries

Schemas

Settings

Connections A connection contains the properties needed to connect to your data.

Add connection

Test connection

Action ▾

☐ Name

Type

Date created



You don't have any connections yet.

Add connection

- Click on Add connection button
- Add a unique and relevant connection name
- Select **Amazon RDS** as Connection Type
- Select **PostgreSQL** as Database engine
- Click Next button

Add connection

☒ Connection properties

☐ Connection access

☐ Review all steps

Set up your connection's properties.

For more information, see [Working with Connections](#).

Connection name

postgres-db-instance

Connection type

Amazon RDS ▾

Database engine

PostgreSQL ▾

☐ Require SSL connection

When selected, connection fails if unable to connect over SSL.

Description (optional)

- Select the instance (the one you have created in RDS)
- Other fields will be filled automatically
- Enter the password
- Click Next button

Add connection

✓ Connection properties

postgres-db-instance
Type: PostgreSQL

○ Connection access

○ Review all steps

Set up access to your data store.

For more information, see [Working with Connections](#).

Instance

postgres-db-instance

Database name

panworld_db

Username

postgresadm

Password

.....

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Next

- Review all parameters and click Finish
- Your connection details are added successfully
- Now goto AWS VPC service and create an Endpoint with service ending with S3 and of type Gateway for your region
- Now you can select this newly created connection and click on Test Connection button to check your connection

1.3 Creating crawler to store schema and metadata information of PostgreSQL database tables:

AWS Glue

Data catalog
Databases
Tables
Connections
Crawlers
Classifiers
Schema registries
Schemas
Settings

Crawlers
A crawler connects to a data store, progresses through a prioritized list of classifiers to determine the schema and metadata tables in your data catalog.

Add crawler

Run crawler

Action ▾

Filter by tags and attributes

☐	Name	Schedule	Status	Logs	Last runtime
 You don't have any crawlers yet. Add crawler					

FTI

- Click on Add crawler button
- Enter a unique and relevant name for the crawler and click Next

Add crawler

- ☒ Crawler info
 - postgres-db-crawler
- ☒ Crawler source type
 - Data stores
- ☐ Data store
- ☐ IAM Role
- ☐ Schedule
- ☐ Output
- ☐ Review all steps

Add information about your crawler

Crawler name

postgres-db-crawler

► Tags, description, security configuration, and classifiers (optional)

Next

- Continue with default options and click Next

Add crawler

- ☒ Crawler info
 - postgres-db-crawler
- ☒ Crawler source type
 - Data stores
- ☐ Data store
- ☐ IAM Role
- ☐ Schedule
- ☐ Output
- ☐ Review all steps

Specify crawler source type

Choose Existing catalog tables to specify catalog tables as the crawler source. The selected tables specify the data stores to crawl. This option doesn't support JDBC data stores.

Crawler source type

- ☒ Data stores
- ☐ Existing catalog tables

Repeat crawls of S3 data stores

- ☒ Crawl all folders
 - Crawl all folders again with every subsequent crawl.
- ☐ Crawl new folders only
 - Only Amazon S3 folders that were added since the last crawl will be crawled. If the schemas are compatible, new partitions will be added to existing tables.
- ☐ Crawl changed folders identified by Amazon S3 Event Notifications
 - Relv on Amazon S3 events to control what folders to crawl.

- Select JDBC as data store
- Select the connection you created in step 1.2 as Connection
- Include path as <PostgreSQL database name>/<PostgreSQL database Schema>/%
- Click Next button
- Continue with default option for Add another data store as No and click Next button

Add crawler

- ✓ Crawler info
 - postgres-db-crawler
- ✓ Crawler source type
 - Data stores
- Data store
- IAM Role
- Schedule
- Output
- Review all steps

Add a data store

Choose a data store

JDBC

Connection

postgres-db-instance

[Add connection](#)

Include path

panworld_db/public/%

You can substitute the percent (%) character for a schema or table. For databases that support schemas, enter MyDatabase/MySchema/% to match all tables in MySchema within MyDatabase. Oracle Database and MySQL don't support schema in the path; instead, enter MyDatabase/%. For Oracle Database, MyDatabase is the system identifier (SID). For information about which JDBC data stores support schema in the path, see [Crawler Properties](#).

- Select the existing IAM role (the one with access to RDS services also)
- Click Next

Add crawler

- ✓ Crawler info
 - postgres-db-crawler
- ✓ Crawler source type
 - Data stores
- ✓ Data store
 - JDBC:postgres-db-i...
- IAM Role
 - arn:aws:iam::406006970390:role/AWSGlueServiceNewRole
- Schedule
- Output

Choose an IAM role

The IAM role allows the crawler to run and access your Amazon S3 data stores. [Learn more](#)

IAM role ⓘ

AWSGlueServiceNewRole



This role must provide permissions similar to the AWS managed policy, **AWSGlueServiceRole**, plus access to your data stores.

You can also create an IAM role on the [IAM console](#).

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[Next](#)

- Continue with the default selection for Frequency as Run on Demand
- Click Next

Add crawler

- ✓ Crawler info
 - postgres-db-crawler
- ✓ Crawler source type
 - Data stores
- ✓ Data store
 - JDBC:postgres-db-i...
- ✓ IAM Role
 - arn:aws:iam::4060069

Create a schedule for this crawler

Frequency

Run on demand

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[Next](#)

- Select the database we created in step 1.1
- Click Next

Add crawler

- ✓ Crawler info
postgres-db-crawler
- ✓ Crawler source type
Data stores
- ✓ Data store
JDBC:postgres-db-i...
- ✓ IAM Role
arn:aws:iam::4060069
70390:role/AWSGlue
ServiceNewRole
- ✓ Schedule
Run on demand
- Output

Configure the crawler's output

Database ⓘ

panworld_db

Add database

Prefix added to tables (optional) ⓘ

Type a prefix added to table names

- Grouping behavior for S3 data (optional)
- Configuration options (optional)

- Review all the parameters and click Finish
- Now select the crawler and click on Run crawler (Make sure you have tables already created in your RDS PostgreSQL database)